BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Algorithmic Thinking

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is decomposition?

2. Which is an OS file-management task?
   A) Adding two scores
   B) Renaming a folder
   C) Routing a packet
   D) Decoding an instruction

3. "Lossless compression deliberately removes arbitrary source-code characters." Correct this. Explain why.

4. A computer needs maintenance and data protection. "Utilities are only apps for writing essays." Correct this. What do utilities do?

Today's objectives

Define an algorithm as an ordered set of steps. Put instructions in the correct logical order.

Key vocabulary - Read, cover, recall

Algorithm
Finite sequence of clear instructions to solve a problem.

Algorithmic thinking
Creating an ordered, logical method for a task.

Sequence
Instructions carried out in their specified order.

Input
Data supplied to an algorithm.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

What is an algorithm?

An algorithm is a finite sequence of clear, ordered instructions that solves a problem. The steps must be sufficiently precise and unambiguous to follow.

Find inputs, actions and result

Read the task and identify what information is supplied, what calculations or decisions happen and what output should be produced.

Order matters

A program must acquire values before using them. Calculating a total before defining item prices is a logic error in the proposed sequence.

Test using an example

Dry-run the steps with example data, record intermediate values and compare the result with what should happen. Correct and refine any unclear step.

I Do - Worked model

A program must input two prices, add them and display the total. Write FOUR ordered algorithm steps and explain why the order matters. [4 marks]

- Input/read the first price [1].
- Input/read the second price [1].
- Calculate the sum from the two values [1].
- Display the total [1]. The calculation must follow both inputs; otherwise required values are unavailable.

We Do - Guided practice

1. A taxi journey fare is a fixed £3 plus £2 per kilometre. Write four ordered steps to calculate and display the fare. [4 marks]

2. A score program adds a 5-point bonus to an entered score. Explain why the input must happen before calculating the total. [4 marks]

3. A pupil writes an algorithm for the area of a rectangle. Explain two criteria that make its steps effective. [4 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the free storage after 7 MB is used.

```
startMB = 20
usedMB = 7
freeMB = startMB - usedMB
print(freeMB)
```

You Do - Independent exam questions

Q2. State what an algorithm is. [2 marks]

Q3. A café calculates the cost of two drinks then adds a £1 service charge. Write FOUR logical steps for the algorithm. [4 marks]

Q4. Explain why clear order, unambiguous steps and testing with sample inputs matter when producing an algorithm. [6 marks]

Exit Ticket - True or False?

An algorithm should give a clear sequence of instructions that can be followed to solve a problem.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Breaking a complex problem into smaller manageable subproblems.

2. B - Renaming a folder

3. Lossless compression restores the original exactly. Losing source-code characters could break the program.

4. Utilities maintain or protect a computer, adding tools beyond the OS. Writing an essay is an application task.

I Do - Explain the model

Input/read the first price [1].

Input/read the second price [1].

Calculate the sum from the two values [1].

Display the total [1]. The calculation must follow both inputs; otherwise required values are unavailable.

We Do - Model answers

1. 1. Input distance in kilometres [1]. 2. Set base fare = 3 [1]. 3. Calculate fare = 3 + 2 x distance [1]. 4. Output fare [1].

2. The input supplies the original score [1], which is needed for the calculation [1]. The program then adds 5 [1] before displaying the updated result [1].

3. The steps must be clear and unambiguous [1], so another person can follow them accurately [1]. They must be in the correct order [1], so length/width are known before calculating and displaying area [1].

Code First - Answer

```
startMB = 20
usedMB = 7
freeMB = startMB - usedMB
print(freeMB)
```

Expected output: 13

You Do - Self-mark

Q2: A set/sequence of clear instructions [1] for solving a problem [1].

Q3: Input price of drink 1 [1]; input drink 2 price [1]; total = price1 + price2 + 1 [1]; output total [1].

Q4: Order ensures inputs exist before later calculations [2]. Unambiguous steps specify actions clearly [2], reducing misinterpretation. Testing with sample inputs [1] allows predicted and actual outcomes to be compared and errors corrected [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.